



Seeing beyond

3D Fluorescence Imaging with Structured Illumination

Optical Sectioning in Widefield Fluorescence Microscopy with ZEISS Apotome

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3 - dimensional fluorescence often produces images with less clarity due to out-of-focus light, particularly in thick specimens. Structured Illumination Microscopy (SIM) provides an effective optical sectioning method that enhances image contrast and resolution by eliminating the unwanted background fluorescence. This application explores the principles, optimisation strategies and advantages of ZEISS Apotome imaging, providing practical insights into grid selection, phase acquisition and the benefits of combining SIM with deconvolution for advanced biological imaging.

Keywords or phrases: Structured Illumination Microscopy (SIM), optical sectioning, 3D fluorescence imaging, grid, contrast, deconvolution, widefield microscopy

Introduction

In conventional widefield fluorescence microscopy an image contains light collected from the structures in the plane of focus as well as light from structures above and below this plane. The light from structures above and below becomes visible in the focal plane because of diffraction and interference and creates a blur in the image.

Diffraction and interference are physical phenomena where light waves interact with each other causing a blurring of the object of interest.

Consequently, a point object which is smaller than the resolution limit of the objective when imaged in 3D is not imaged as a point but as an hourglass shape known as Point Spread Function (PSF). When studying thick specimen, this out-of-focus light degrades the image quality by increasing the background fluorescence in the

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image and therefore decrease the signal-to-noise ratio (SNR) and the image contrast.

Such images are often not suitable for reasonable analysis and interpretation of the 3-dimensional structure of the objects.

Therefore, the motivation is to remove the out-of-focus light and obtain images of the objects of interest with highest contrast and highest resolution. This is the principle in optical sectioning. Besides the predominant confocal method using a pinhole in an optically conjugate plane, other approaches have been developed to obtain optical sectioning, including Structured Illumination.

As mentioned above, the main motivation of optical sectioning is improving the contrast by removing the out-of-focus light. This usually comes at the expense of decreasing the overall signal. When imaging weakly fluorescing samples this can be overcome by increasing the thickness of the optical section or by increasing the acquisition time. The latter has the negative side effect of putting higher burden on the sample leading to bleaching and increased phototoxicity, whereas the former increases the SNR. So, the major trade-off when doing optical imaging is the decrease in signal, and the skill of the operator to define the compromise between signal, resolution (i.e. thickness), photodamage and acquisition time. The thickness of the optical section is determined differently dependent on the sectioning method applied.

For instance, in confocal scanning microscopy, the size of the pinhole determines the thickness of the optical section. Increasing the pinhole will increase the thickness of the optical sample and increase the overall signal. The optimal thickness of the optical section is also determined by the numerical aperture (NA) of the objective lens, where high NA allows for thinner optical sectioning. In structured illumination, the thickness of the optical section can be altered by changing the frequency of the grid pattern. This also means that different objectives lenses with various numerical

apertures require different grid pattern frequencies.

The principle in structured illumination used in Apotome, is based on projection of a defined grid into the focal plane of a widefield fluorescence microscope. As the grid is moving in a lateral manner, a defined number of images, termed phases, are captured by a camera. The light emitted by structures, which are in the focal plane, will vary strongly in intensities when the illuminating grid lines move across them. If structures are not in focus, the wandering grid lines will not cause the emitted light to fluctuate as strongly.

By calculating the difference between the different phase images, it is possible to discriminate between in-focus and out-of-focus light. The differences in intensities are calculated pixel-by-pixel and the final image contains only the in-focus information which is an optical section.

Optimal Choice of Grid Pattern

Apotome is equipped with a motorized grid changer, which allows to select between three grid patterns: Low, Medium, and High (Figure 1).

The grid patterns in Apotome are different for Axio Imager.2, Axio Observer, and Axio Zoom.V16, which is due to differences in the optical path between the three stands. The grid pattern determines the thickness of the optical section (box 1).

This application focuses on Axio Observer, but the principles are the same for all stands.



Figure 1: Three grid patterns available in ZEISS Apotome. The change between the pattern is controlled by the software (or the user) and is optimized according to the selected objective.

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Fluorescence image stacks of a 1 μm green, fluorescent beads on Axio Observer were acquired using the three grid patterns.

As shown in figure 2, the different grid patterns decreases the size of the point spread function (PSF) independently of the objective. As shown in the top panel for the Plan- APOCHROMAT 20x / 0.8 objective (compare Figure 2, a – d), the axial expansion of the PSF is clearly reduced when changing the grid pattern. Figure 2a shows the XZ plane of the PSF when observed in widefield microscope, figures 2b–2d show the axial expansion for the High, Medium, and Low grid pattern respectively.

Also, the PSF size observed using a Plan-APOCHROMAT objective 40x / 1.4 is reduced in combination with Apotome (Figure 2, lower panel).

When comparing the PSF acquired using widefield and high grid, a significant reduction is observed (compare Figure 2, a' and b') but in contrast to the Plan- APOCHROMAT objective 20x / 0.8, the PSF of Plan-APOCHROMAT 40x / 1.4 is only reduced minimally when changing the grid pattern from Medium to Low (compare Figures 2, c' and d').

This demonstrates the rationale for the automatic grid pattern selected by the software and highlights the importance of the different grid patterns. The optimal choice of grid pattern is achieved, when the diffracted light above and below the core of the PSF is effectively removed but without making the optical section too thin. This would result in an increase in signal-to-background, which in return would reduce the contrast. As shown, the low grid pattern is the optimal choice for Plan-APOCHROMAT 20x / 0.8, whereas for Plan-APOCHROMAT 40x / 1.4 the Medium grid pattern is the optimal choice. As a rule of thumb, the high magnification and / or high numerical aperture require High / Medium grid patterns, and low magnification and / or low numerical aperture require Medium / Low grid patterns.

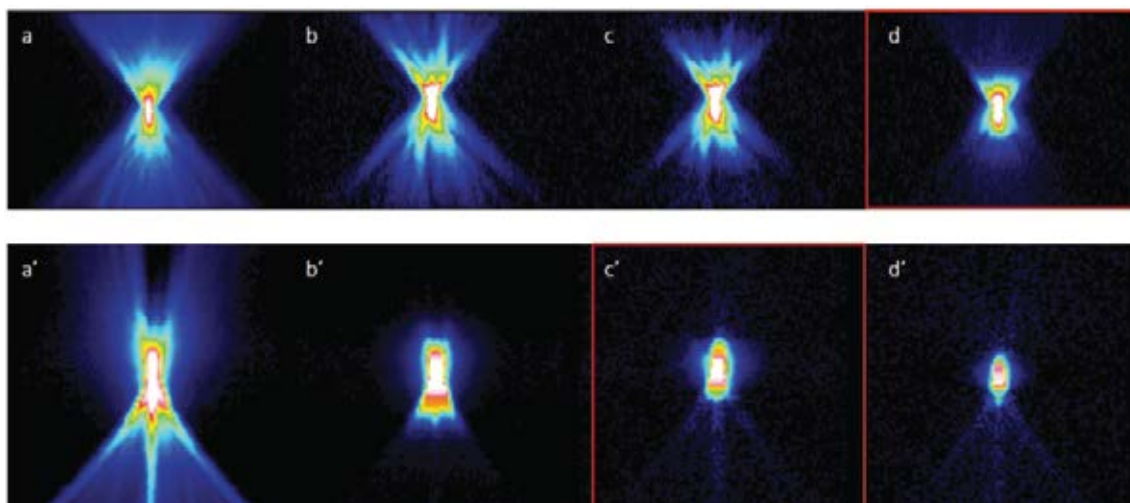


Figure 2: PSF of 1 μm bead using Plan-APOCHROMAT 20x / 0.80 (top panel) and Plan-APOCHROMAT 40x / 1.4 Oil. The XZ view is shown for widefield (a), High (b), Medium (c), and Low (d) grid patterns were used for acquisition. The red box highlights the automatic choice of grid pattern.

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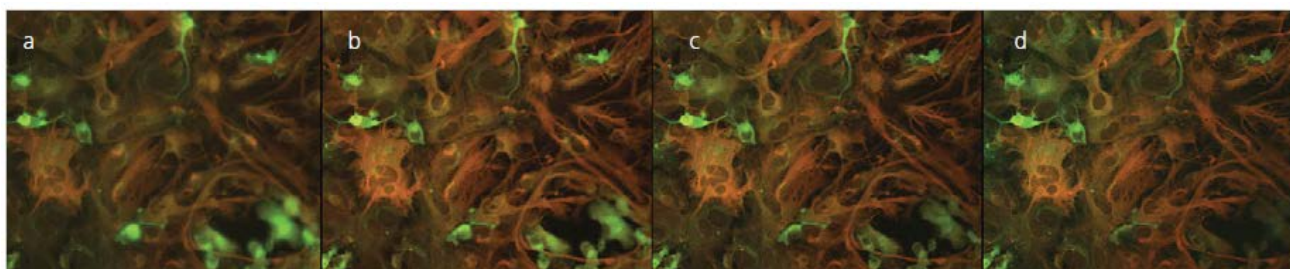


Figure 3a: Grid pattern determines the thickness of the optical section. α -Cathepsin B:Alexa 488 and α -GFAP:Alexa546 staining of astrocytes imaged using Plan-APOCHROMAT 40x / 1.4 Oil with widefield (a) Low (b), Medium (c) and High (d) grid pattern.

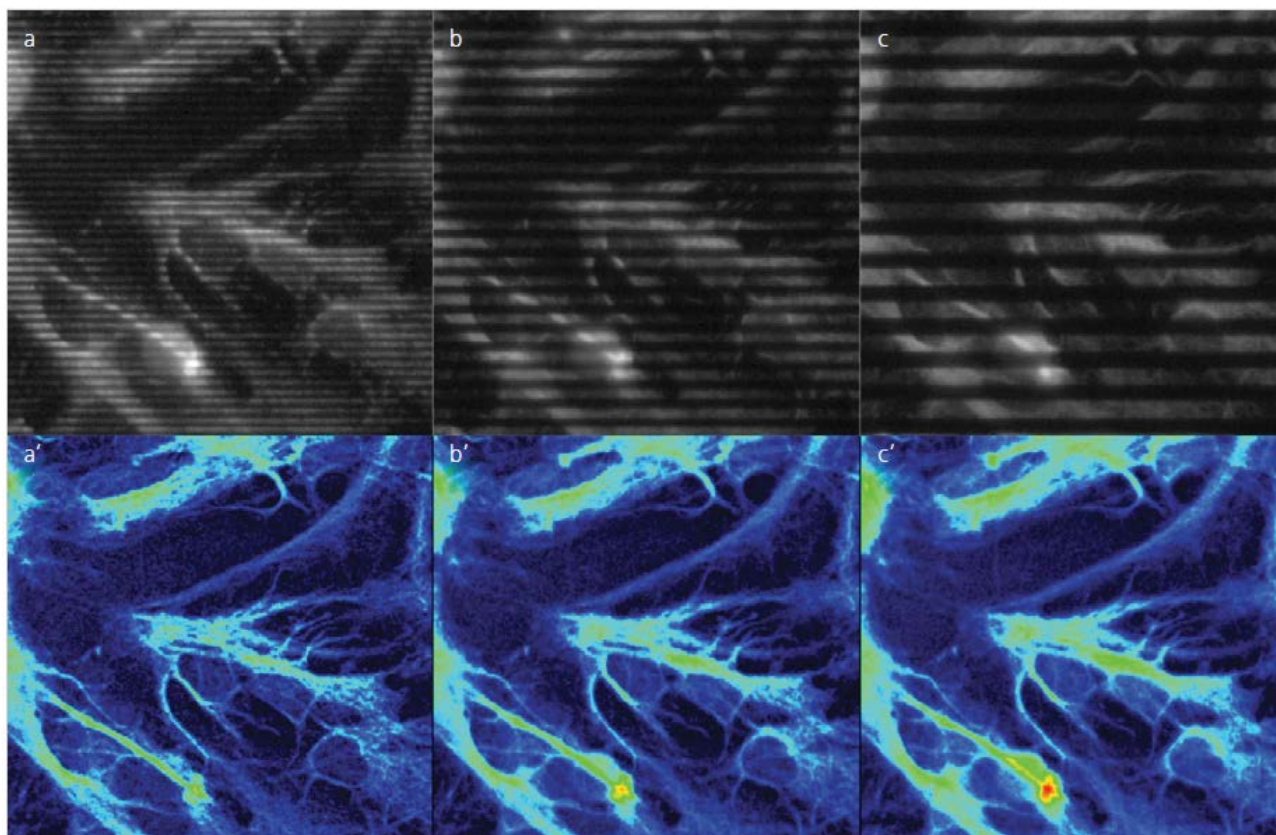


Figure 3b: Grid pattern determines the thickness of the optical section. GFAP:Alexa545 staining of astrocytes imaged using Plan-APOCHROMAT 40x / 1.4 Oil with Low (a, a'), Medium (b, b') and High (c, c') grid pattern. In upper panel (a – c) the raw images with the projected grid pattern are shown. The upper row shows raw images with grid lines, the lower row shows corresponding optical sections with identical settings for the look up table (LUT). The lower images show the optical sections derived from.

Thus, the grid pattern alters the PSF, which consequently determines the thickness of the optical section. This flexibility can be used beneficially by the user to increase the signal in weak fluorescent samples by giving up some resolution or increase the contrast in bright samples by deliberately choosing a lower or

higher grid pattern respectively.

As shown in figure 3, using the grid pattern increases the signal-to-background compared to the widefield settings (compare 3a – widefield and 3b – Apotome). Moreover, when changing the grid pattern from

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Medium to Low (compare figure 3c and 3d), the contrast is noticeably enhanced, and the section becomes thinner. This become more obvious when viewing a single channel only (compare Fig. 3a' – d').

The Number of Phases

The grid is shifted by a given number of phases in such fashion that all in-focus structures in the sample become covered once by the moving grid line. Each phase image is captured as an individual image prominently showing the grid lines.

The resulting CZI image contains all phase images and is called Apotome raw image The optical section is then processed from the raw images online when viewing the

image in the ZEN imaging software. The number of phases / images can be selected in ZEN and varies from 3 to 15 with 5 phases being the default. The number of phases has little to no effect on the size of the PSF but has a marked influence on the quality of the processed image. As shown in figure 4, the XZ view PSF of a 1 μm bead is unaltered from images captured using 3 (a), 5 (b) and 10 (c) phases. However, as structured illumination is based on subtraction of consecutive images, this technique is sensitive to bleaching and vibration especially when occurring during capture of individual phase images. This has been addressed in the ZEN imaging software with advanced correction algorithms for bleaching and minor off-grid positions. These corrections remove any residual grid lines effectively. In extreme cases images can still be salvaged by employing an additional Fourier filter which stamps out grid lines.

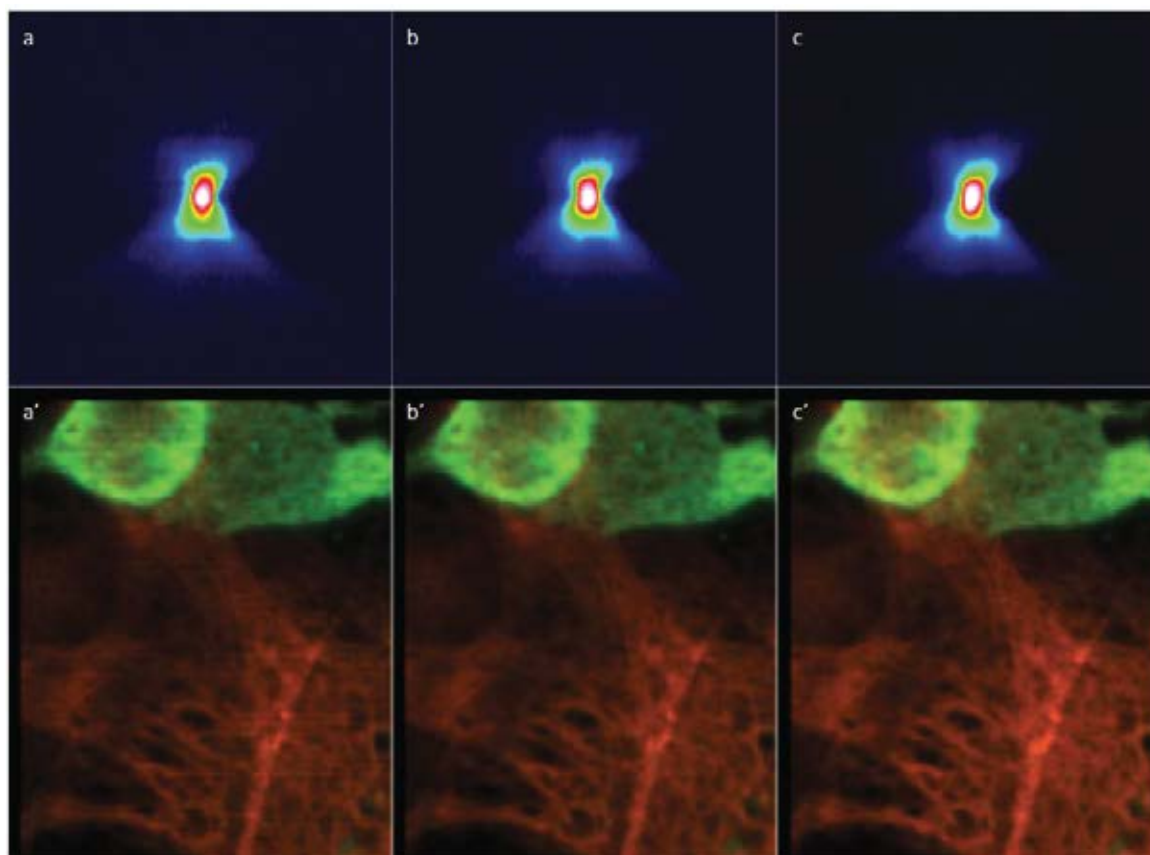


Figure 4: Number of phases determines the signal-to-background and has no impact on the PSF. XZ view of a 1 μm bead (a – c) and close-up detail of astrocytes stained with α -Cathepsin B:Alexa 488 and α - FAP:Alexa546 (a' – c'). The images were captured using 3 phases (a, a'), 5 phases (b, b'), and 10 phases (c, c'), and captured using Plan-APOCHROMAT 40x/1.4 Oil.

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An increase in the number of phases is a good practice for three reasons: first, it serves to decrease camera noise due to averaging. Secondly, it minimizes the occurrence of residual stripes in the processed image. Thirdly, and probably most importantly, an increase in the number of phase images can increase the resolution when combined with Deconvolution.

images processed from 5 (b') and 10 phases (c'). Furthermore, the signal-to-background ratio is improved with the number of phases (compare c' to b' / a') due to added benefit of averaging.

Apotome raw images can be subjected to a regularized inverse filter 3D-deconvolution.

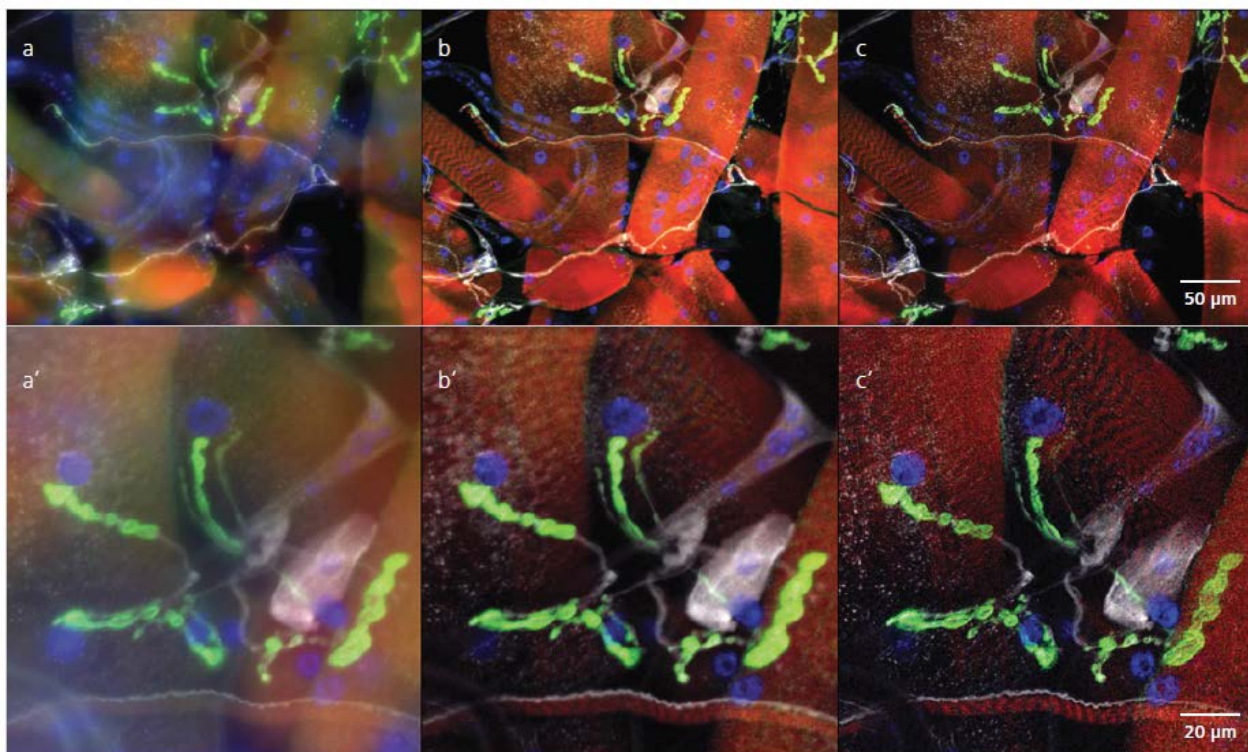


Figure 5: Deconvolution in combination with Apotome improves spatial resolution and restores information. *Drosophila melanogaster* neuromuscular junctions, acquired with Plan-NEOFLUAR 40x/0.75 objective. 6 phases were acquired. Green (Alexa 488): synapses, Red (Alexa 568): muscle; White (Alexa 647): motor neurons; Blue (Hoechst 33258): nucleus. Courtesy of K. V. Raghavan, National Centre for Biological Sciences, Bangalore, Republic of India. Shown in the upper panel is a maximum intensity projection of the z-stack processed as widefield (a), sectioned (b) and sectioned/deconvolved (c), scalebar shows 50 μ m. The lower panel shows a larger magnification of a single neuromuscular junction in widefield (a'), sectioned (b') and sectioned/deconvolved (c') processing, scalebar shows 20 μ m.

ZEISS Apotome and Deconvolution

The decision to preserve the phase images when capturing images using Apotome has an important consequence:

As shown in figure 4 (lower panel) the number of phases improves the signal and minimizes residual stripes:

processing images from 3 phases only, residual stripes are quite apparent in the image (a') when compared to

While 3D-deconvolution can in principle be applied to all fluorescence images, either widefield or confocal, if you have knowledge about the PSF of the acquisition conditions, the combination of structured illumination and deconvolution provides some unique advantages to classical deconvolution. In widefield deconvolution the PSF is fairly large and in order to reconstruct 3D information, iterative processing steps have to be employed to achieve at least a partially confocal PSF.

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The major advantage of this is, that all light emanating from the sample is being used making widefield deconvolution a very sensitive method for 3D imaging. In confocal microscopy, due to the optical sectioning capability of the microscope, 3D information is inherently captured at the expense of throwing away information which has been generated by the sample. Deconvolving confocal datasets can improve contrast and remove noise but can never restore the information lost during image capture.

Apotome deconvolution can do both: keeping the widefield information but at the same time creating optical sections via structured illumination. Signal is therefore restored based on the deconvolved widefield information and when combined with the optical sectioning properties of Apotome can lead to higher resolution results than either method alone could.

Figure 5 shows images of a drosophila neuromuscular junction section acquired with Apotome.2. Structures in the widefield image (a, a') are severely blurred and hard to see.

Adding standard Apotome processing already clears up the blur and reveals the structures (b, b'). However, the finer structures of the synapses e.g. only become visible in the deconvolved version of the image (c, c').

Due to the already confocal nature of the structured illumination, it is not necessary to employ iterative deconvolution algorithms. The rapid processing possible with the regularized inverse filter deconvolution provides best results while only adding moderate processing times making this combination even more attractive.

Conclusion

Optical sectioning using Apotome with ZEN provides features that were previously not well explained, particularly the variable grid pattern, phases, correction algorithms and the combination with 3D-deconvolution.

The choice of grid pattern determines the thickness of the optical section. The user can take advantage of the different grid pattern for instance in order to increase the signal for samples with low signal (thick sectioning) or increase the signal-to-background (thin sectioning). The optimal grid pattern is determined by the software and is dependent on the objective and the excitation wavelength.

The number of phases increase the signal-to-background by a process similar to averaging of the images. Averaging reduces the random shot noise in the final image by calculating an average of each pixel from multiple images. As shown here, the number of phases do little to nothing against the thickness of the optical section. The optimal choice of the number of phases is a compromise between acquisition time and the expected signal-to-background in the final image.

When acquiring few phases (< 5), the processed images are prone to stripe artefacts. Acquiring significantly more phases than 5 will bleach the sample more at proportionally less benefit for image quality. Zeiss' recommendation is to use the default 5 phase setting as the low number of images causes little bleaching damage to the sample and the acquisition time is reasonable.

Deconvolution can increase resolution and restore the lower dynamic range of images after Apotome processing.

In Apotome processing the out-of-focus light is subtracted from the image containing both the focus and the out-of-focus light. This generates a final image with low signal.

Deconvolution increases the signal in the final image due to reassigning the out-of-focus to the focus plane and can enhance the resolution beyond the resolution limit.

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Materials List and Image Acquisition

1) Samples:

- 1 µm green / yellow fluorescent beads
- Astrocytes stained with α-GFAP (Alexa 546), and α-Cathepsin B (Alexa 488)

2) **Stand:** Axio Observer.Z1 configured for standard fluorescence microscopy

3) **Light source:** HXP 120

4) Filters:

- FS38 (Alexa 488 and green, fluorescent beads)
- FS43 (Alexa 546).

5) Objectives:

- Plan-APOCHROMAT 20x / 0.8
- Plan-APOCHROMAT 40x / 1.4 Oil

6) Channels were configured according to Smart Setup

7) 20 – 30 µm z-stacks were acquired using optimal settings

8) Neuromuscular junctions' section, NCBS Bangalore, India



ZEISS Apotome

Section Thickness

The section thickness is specified by the grid pattern.

$$\text{section thickness} = \frac{3.83}{16\pi} \times \frac{\lambda \times 10^{-3}}{n \times \sin^2(a/2) \times v \times (1-v/2)}$$

$$v = \frac{M \times \lambda \times f \times 10^6}{m \times NA}$$

M = objective magnification

n = refractive index of the immersion media

m = correction factor for stand

(1.3 for Axio Imager and 2.57 for Axio Observer)

λ = wavelength in nm

f = grid constant in lines per mm

As shown from the formula, the thickness of the optical section is determined by the objective in use (i.e. magnification, numerical aperture, immersion media), the excitation wavelength, the microscope stand, and the grid.

The ZEN imaging software will automatically calculate the theoretical section thickness based on the configuration in use, and the value is shown in the software. The automatic grid selection aims to keep the section thickness in proximity to one Rayleigh Unit, d :

$$d = \frac{2 \times \lambda \times n}{(NA)^2}$$

Example: Axio Observer equipped with Plan-APOCHROMAT 20x / 0.8 has a calculated Rayleigh Unit and section thickness as follows. The automatic grid selection is indicated in bold.

High grid = 3.2 RU / 4.9 µm, Medium grid = 1.8 RU / 2.8 µm, and Low grid = **1.0 RU / 1.5 µm**.

Axio Observer equipped with Plan-APOCHROMAT 40x / 1.4 Oil has a calculated section thickness as follows:

High grid = 2.4 RU / 1.8 µm, **Medium grid = 1.4 RU / 1.0 µm**, and Low grid = 0.7 RU / 0.6 µm.